# NARAYANA ENGINEERING COLLEGE :: GUDUR

## DEPARTMENT OF ECE

| CourseDetails | | | | | | | |
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| **Class** | : | **II -B.Tech** | **Semester** | | **II** | **Year** | : **2024-25** |
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| **Regulation** | : | **NECR-23** | **Faculty** | | **:Dr. B. Malakonda Reddy** | | |

**Question Bank /Topic Learning Outcomes**

(Table format as given below is suitable for Theory, MCQ, Lab, and Tutorial. Columns can be added or deleted. No need to take printouts)

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| 1 | **Derive wave equation for conducting medium in terms of Electric field intensity and Electric flux Density.**<br>Consider Maxwell's Equation expressed in $\bar{E}$ and $\bar{H}$ from Faraday's Law<br>$$\nabla \times \bar{\mathbf{E}} = -\frac{\partial \bar{\mathbf{B}}}{\partial t}$$<br>Take Curl both sides |

$$\therefore \quad \nabla \times \nabla \times \overline{E} = -\mu\left[\frac{\partial}{\partial t}(\nabla \times \overline{H})\right] \quad \text{... (2)}$$

Using vector identity to the left of equation (2),

$$\therefore \quad \nabla(\nabla \cdot \overline{E}) - \nabla^2 \overline{E} = -\mu\left[\frac{\partial}{\partial t}(\nabla \times \overline{H})\right] \quad \text{... (3)}$$

But according to another Maxwell's equation,

$$\nabla \times \overline{H} = \overline{J} + \frac{\partial \overline{D}}{\partial t}$$

Putting value of $\nabla \times \overline{H}$ in equation (3),

$$\therefore \quad \nabla(\nabla \cdot \overline{E}) - \nabla^2 \overline{E} = -\mu\left[\frac{\partial}{\partial t}\left(\overline{J} + \frac{\partial \overline{D}}{\partial t}\right)\right] \quad \text{... (4)}$$

Since most of the regions are source or charge free,

$$\therefore \quad \nabla \cdot \overline{E} = 0$$

$$\therefore \quad \nabla(\nabla \cdot \overline{E}) = 0$$

Putting value of $\nabla(\nabla \cdot \overline{E})$ in equation (4), assuming charge free medium,

$$-\nabla^2 \overline{E} = -\mu\left[\frac{\partial}{\partial t}\left(\overline{J} + \frac{\partial \overline{D}}{\partial t}\right)\right]$$

Making both sides positive,

$$\nabla^2 \overline{E} = \mu\left[\frac{\partial}{\partial t}\left(\overline{J} + \frac{\partial \overline{D}}{\partial t}\right)\right] \quad \text{... (5)}$$

Consider a general electromagnetic wave with both the fields, $\overline{E}$ and $\overline{H}$ varying with respect to time. When any field varies with respect to time, its partial derivative taken with respect to time can be replaced by $j\omega$ Rewriting equation (5) in phasor form,

$$\therefore \quad \nabla^2 \overline{E} = \mu\left[j\omega\left(\overline{J} + j\omega \overline{D}\right)\right]$$

$$\therefore \quad \nabla^2 \overline{E} = j\omega\mu\left[(\sigma \overline{E}) + j\omega(\varepsilon \overline{E})\right]$$

$$\therefore \quad \nabla^2 \overline{E} = \left[j\omega\sigma\mu \overline{E} + (j\omega)\varepsilon\mu \overline{E}\right]$$

$$\therefore \quad \nabla^2 \overline{E} = \left[j\omega\mu(\sigma + j\omega\varepsilon)\right]\overline{E} \quad \text{... (6)}$$

The above equation in represents wave equation for conducting medium in terms of Electric field intensity.

$$\nabla^2 \overline{D} = [j\omega\mu(\sigma + j\omega\varepsilon)]\overline{D}$$

The above equation in represents wave equation for conducting medium in terms of Electric flux Density.

2 **Derive an expression for wave propagation in good dielectric.**

for practical dielectric, conductivity is not zero ; but it is very small. The condition for practical dielectric is that the loss in the signal is very small i.e. $\sigma << j\omega\varepsilon$.

The propagation constant is given by

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}$$

$$\therefore \quad \gamma = \sqrt{(j\omega\mu)(j\omega\varepsilon)\left(1 + \frac{\sigma}{j\omega\varepsilon}\right)}$$

$$\therefore \quad \gamma = j\,\omega\sqrt{\mu\varepsilon}\,\sqrt{1 + \frac{\sigma}{j\omega\varepsilon}}$$

Consider radical term. Mathematically using binomial theorem,

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots$$

where $|x| < 1$.

In our case $|x| < 1$ and $n = \frac{1}{2}$, then neglecting higher order terms, we can write,

$$\left(1+\frac{\sigma}{j\omega\varepsilon}\right)^{\frac{1}{2}} = 1 + \frac{1}{2}\left(\frac{\sigma}{j\omega\varepsilon}\right)$$

Substituting above value for the radical term,

$$\gamma = j\omega\sqrt{\mu\varepsilon}\left[1+\frac{\sigma}{j2\omega\varepsilon}\right]$$

$$\therefore \quad \gamma = \alpha + j\beta = \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}} + j\omega\sqrt{\mu\varepsilon} \qquad \ldots (1)$$

Comparing the real terms, the attenuation constant is given by

$$\alpha = \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}} \quad \text{Np/m} \qquad \ldots (2)$$

Similarly comparing the imaginary terms, the phase constant is given by

$$\beta = \omega\sqrt{\mu\varepsilon} \quad \text{rad/m} \qquad \ldots (3)$$

In general, the intrinsic impedance $\eta$ is given by,

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = \sqrt{\frac{j\omega\mu}{j\omega\varepsilon\left(1+\frac{\sigma}{j\omega\varepsilon}\right)}}$$

$$\therefore \quad \eta = \sqrt{\frac{\mu}{\varepsilon}}\sqrt{\frac{1}{1+\frac{\sigma}{j\omega\varepsilon}}}$$

$$\therefore \quad \eta = \sqrt{\frac{\mu}{\varepsilon}}\left[1+\frac{\sigma}{j\omega\varepsilon}\right]^{-\frac{1}{2}}$$

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| | Using binomial theorem,<br>$$(1+x)^{-n} = 1 - nx + \frac{n(n-1)}{2!}x^2 - \frac{n(n-1)(n-2)}{3!}x^3 \ldots$$<br>As $x = \frac{\sigma}{j\omega\varepsilon}$ is very small as compared to 1, so neglecting higher order terms,<br>$$(1+x)^{-n} = 1 - nx$$<br>Using above result, the intrinsic impedance can be written as,<br>$$\eta = \sqrt{\frac{\mu}{\varepsilon}}\left[1 - \frac{1}{2}\left(\frac{\sigma}{j\omega\varepsilon}\right)\right]$$<br>$$\therefore \quad \eta = \sqrt{\frac{\mu}{\varepsilon}}\left(1 + j\frac{\sigma}{j\omega\varepsilon}\right)\Omega \quad \ldots (4)$$ |
| 3 | **In a nonmagnetic material H= 30cos(2Π x $10^8$t-6x)$a_y$ mA/m. Calculate:**<br>**(a) The intrinsic impedance.**<br>**(b) The Poynting vector.**<br>**(c) The time average power crossing the surface. x = 1, 0 < y < 2, 0 < z < 3m.**<br>Given data<br>H= 30cos(2Π x $10^8$t-6x)$a_y$mA/m<br>H= 30X$10^{-3}$cos(2Π x $10^8$t-6x)$a_y$A/m<br>$\omega = 2\Pi \times 10^8$<br>$\beta = 6$<br>$H_m = 30X10^{-3}$<br>For non magnetic material $\mu=\mu_0$ and $\varepsilon=\varepsilon_0\varepsilon_r$<br>$$\beta = \omega\sqrt{\mu\varepsilon}$$<br>$$6 = 2\pi X10^8\sqrt{\mu_0\varepsilon_0\varepsilon_r}$$<br>$$6 = \frac{2\pi X10^8}{3X10^8}\sqrt{\varepsilon_r}$$<br>$$\varepsilon_r = 81$$ |

(a) The intrinsic impedance.

$$\eta = \sqrt{\frac{\mu}{\varepsilon}} = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{377}{\sqrt{81}} = 41.88\Omega$$

b) The Poynting vector.

$$\overline{P} = \overline{E} X \overline{H}$$

$$\overline{P} = \eta H^2 \overline{a}_x$$

$$\overline{P} = 41.88 X (30X10^{-3})^2 \cos^2(2\pi \text{ x } 10^8 \text{t - 6x}) \overline{a}_x$$

$$\overline{P} = P\overline{a}_x$$

$$\overline{P}_{avg} = \frac{1}{T}\int_0^T P dt \overline{a}_x$$

Substitute P in the integral

c) The time average power crossing the surface. x = 1, 0 < y < 2, 0 < z < 3m.

$$P_{avg} = \oint_s \overline{P}_{avg} . d\overline{S}$$

$$P_{avg} = \iint P.dydz$$

Use these limits: x =

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| 4 | **Derive an expression for wave propagation in perfect dielectric.**<br>The perfect dielectric medium is also calling as<br>•Not free Space but $\sigma = 0$<br>•Loss less Dielectric medium<br>•Non conducting medium<br>In perfect dielectric medium, the conductivity is zero i.e., $\sigma = 0$<br>In perfect dielectric medium, $\sigma = 0$, $\varepsilon = \varepsilon_0 \varepsilon_r$ ; $\mu = \mu_0$<br>If the wave is propagating in perfect dielectric medium,<br>The propagation constant $\gamma$ is given by,<br>$\gamma = \sqrt{j\omega\mu(0+j\omega\varepsilon)} = \pm j\omega\sqrt{\mu\varepsilon}$<br>$\therefore \quad \gamma = \alpha + j\beta = \pm j\omega\sqrt{\mu\varepsilon}$ ... (1)<br>From equation (1) it is clear that, propagation constant is purely imaginary. It indicates in a perfect dielectric medium, attenuation constant $\alpha$ is zero. Let us select value of $\beta$ which gives propagation of wave in positive z-direction.<br>$\therefore \quad \alpha = 0, \quad \beta = \omega\sqrt{\mu\varepsilon}$ ... (2)<br>Similarly, if the wave is propagating in perfect dielectric medium,<br>$\eta = \sqrt{\frac{j\omega\mu}{0+j\omega\varepsilon}} = \sqrt{\frac{\mu}{\varepsilon}}\ \Omega$ ... (3)<br>As<br>$\eta = \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_0}{\varepsilon_0\varepsilon_r}} = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{377\Omega}{\sqrt{\varepsilon_r}} = \frac{120\pi}{\sqrt{\varepsilon_r}}\Omega$ |

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| | Thus intrinsic impedance $\eta$ is real resistive. That means phase angle of intrinsic impedance is zero. But the phase angle of intrinsic impedance is zero means phase difference between $\overline{E}$ and $\overline{H}$ is zero. In other words, **for a perfect dielectric, both the fields $\overline{E}$ and $\overline{H}$, are in phase.**<br><br>As in perfect dielectric, $\sigma = 0$, attenuation constant $(\alpha)$ is also zero. As wave propagates, only the phase $(\beta)$ changes. Thus no attenuation i.e. $\alpha = 0$ means no loss. So perfect diectric medium is also called **lossless dielectric.**<br><br>The velocity of propagation in the perfect dielectric is given by,<br>$$v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{\omega}{\beta} \quad \text{... (4)}$$<br>If $\lambda$ is the wavelength of one cycle of the propagating wave then velocity is given by<br>$$v = \lambda f \ \text{m/s} \quad \text{... (5)}$$ | |
| 5 | **Obtain the relation between E & H in uniform plane wave and obtain the characteristic impedance of the non conducting medium.**<br>Assume an electromagnetic wave travelling in free space. Consider that an electric field is in x-direction ; while a magnetic field is in y-direction. Both the fields will not vary with x and y ; but with z only. They will also change with time as the wave propagates in free space.<br><br>Consider Maxwell's equation expressed in $\overline{E}$ and $\overline{H}$ as<br>$$\nabla \times \overline{H} = \overline{J} + \frac{\partial \overline{D}}{\partial t}$$<br>Let us assume that a free space is perfect dielectric, then $\overline{J} = 0$,<br>$$\therefore \quad \nabla \times \overline{H} = \frac{\partial \overline{D}}{\partial t}$$ | |

Expressing $\overline{D}$ in rectangular co-ordinate system,

$$\nabla \times \overline{H} = \frac{\partial}{\partial t}\left[D_x \overline{a}_x + D_y \overline{a}_y + D_z \overline{a}_z\right]$$

Writing curl of $\overline{H}$ on left of equation,

$$\left[\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}\right]\overline{a}_x + \left[\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x}\right]\overline{a}_y + \left[\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y}\right]\overline{a}_z$$

$$= \frac{\partial}{\partial t}\left[D_x \overline{a}_x + D_y \overline{a}_y + D_z \overline{a}_z\right]$$

As $\overline{H}$ is in y-direction, $H_x = H_z = 0$,

$$\therefore -\frac{\partial H_y}{\partial z}\overline{a}_x + \frac{\partial H_y}{\partial x}\overline{a}_z = \frac{\partial}{\partial t}\left[D_x \overline{a}_x + D_y \overline{a}_y + D_z \overline{a}_z\right]$$

Also $H_y$ is not changing with x, as it is uniform in x-y plane, so $\frac{\partial H_y}{\partial x} = 0$

$$\therefore \quad -\frac{\partial H_y}{\partial z}\overline{a}_x = \frac{\partial}{\partial t}\left[D_x \overline{a}_x + D_y \overline{a}_y + D_z \overline{a}_z\right]$$

Equating L.H.S. and R.H.S. of above equation directionwise, we can write,

$$-\frac{\partial H_y}{\partial z} = \frac{\partial D_x}{\partial t}$$

$$\therefore \quad -\frac{\partial H_y}{\partial z} = \varepsilon \frac{\partial E_x}{\partial t} \quad .... \quad \because \quad \overline{D} = \varepsilon \overline{E}$$

$$\therefore \quad \frac{\partial H_y}{\partial z} = -\varepsilon \frac{\partial E_x}{\partial t} \qquad ... (1)$$

Now consider Maxwell's equation derived from Faraday's law,

$$\nabla \times \overline{E} = -\frac{\partial \overline{B}}{\partial t}$$

Using rectangular co-ordinate system, we can write,

$$\left[\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y}\right]\overline{a}_x + \left[\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right]\overline{a}_y + \left[\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x}\right]\overline{a}_z$$

$$= -\frac{\partial}{\partial t}\left[B_x \overline{a}_x + B_y \overline{a}_y + B_z \overline{a}_z\right]$$

As $\overline{E}$ is in x-direction, $E_y = E_z = 0$,

$$\therefore \frac{\partial E_x}{\partial z}\overline{a}_y + \frac{\partial E_x}{\partial y}\overline{a}_z = -\frac{\partial}{\partial t}\left[B_x \overline{a}_x + B_y \overline{a}_y + B_z \overline{a}_z\right]$$

Also component $E_x$ is not varying with y, as it is uniform in x-y plane, so $\frac{\partial E_x}{\partial y} = 0$,

$$\therefore \quad \frac{\partial E_x}{\partial z}\overline{a}_y = -\frac{\partial}{\partial t}\left[B_x \overline{a}_x + B_y \overline{a}_y + B_z \overline{a}_z\right]$$

Equating L.H.S. and R.H.S. of above equation directionwise, we can write,

$$\frac{\partial E_x}{\partial z} = -\frac{\partial B_y}{\partial t}$$

$$\therefore \quad \frac{\partial E_x}{\partial z} = -\mu \frac{\partial H_y}{\partial t} \quad \text{.....} \quad \overline{B} = \mu \overline{H}$$

$$\therefore \quad \frac{\partial H_y}{\partial t} = -\frac{1}{\mu}\frac{\partial E_x}{\partial z} \qquad \text{... (2)}$$

Let the wave travelling in the positive z-direction as

$$\boxed{E_x = E_0 e^{j\omega t} e^{-\gamma z}} \qquad \text{Eq ------- (3)}$$

Differentiate Eq., (3) w.r.t z

$$\frac{\partial E_x}{\partial z} = (-\gamma) E_0 e^{j\omega t} e^{-\gamma z}$$ Eq ------- (4)

Substitute Eq., (4) in Eq., (1)

$$(-\gamma) E_0 e^{j\omega t} e^{-\gamma z} = -\mu_0 \frac{\partial H_y}{\partial t}$$

$$\gamma E_0 e^{j\omega t} e^{-\gamma z} = \mu_0 \frac{\partial H_y}{\partial t}$$

$$\frac{\partial H_y}{\partial t} = \frac{\gamma E_0 e^{j\omega t} e^{-\gamma z}}{\mu_0}$$ Eq ------- (5)

Integrate Eq., (5) both sides wr.t 't'

$$H_y = \frac{\gamma}{j\omega\mu_0} (E_0 e^{j\omega t} e^{-\gamma z})$$ Eq ------- (6)

Eq., (6) can be written as

$$H_y = \frac{\gamma E_x}{j\omega\mu_0}$$ from Eq(3)

$$\frac{E_x}{H_y} = \frac{j\omega\mu_0}{\gamma}$$ Eq ------- (7)

For any medium Eq., (7) can be written as

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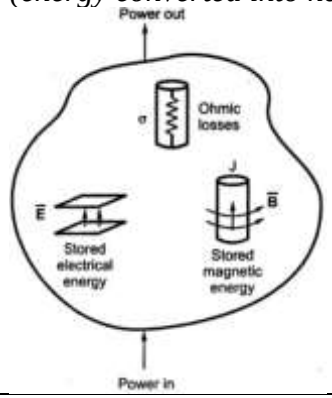
| | $$\frac{E_x}{H_y} = \frac{j\omega\mu}{\gamma} = \frac{j\omega\mu}{\sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}}$$ As we know, the ratio between is E and H is intrinsic impedance $$\eta = \frac{E_x}{H_y} = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}}$$ This gives Relation between E and H in any medium. |
| 6 | **Derive an expression for Poynting's theorem.**<br>Poynting Theorem is based on **Law of Conservation of Energy**<br>Poynting Theorem states that "*the power coming out of the closed volume is equal to the total decrease in EM energy per unit time i e power loss from the*<br>*Volume which constitutes of rate of decrease in magnetic energy stored in the volume rate of decrease in electric energy stored in the volume Ohmic power loss (energy converted into heat energy per unit time) in the volume*"<br>  <br> |

Consider Maxwell's equations as given below :

$$\nabla \times \overline{E} = -\frac{\partial \overline{B}}{\partial t} = -\mu \frac{\partial \overline{H}}{\partial t} \quad \text{... (1)}$$

$$\nabla \times \overline{H} = \overline{J} + \frac{\partial \overline{D}}{\partial t} = \sigma \overline{E} + \varepsilon \frac{\partial \overline{E}}{\partial t} \quad \text{... (2)}$$

Dotting both the sides of equation (3) with $\overline{E}$, we get,

$$\overline{E} \cdot (\nabla \times \overline{H}) = \overline{E} \cdot (\sigma \overline{E}) + \overline{E} \cdot \left( \varepsilon \frac{\partial \overline{E}}{\partial t} \right) \quad \text{... (3)}$$

Let us make use of a vector identity as given below,

$$\nabla \cdot (\overline{A} \times \overline{B}) = \overline{B} \cdot (\nabla \times \overline{A}) - \overline{A} \cdot (\nabla \times \overline{B})$$

Applying above vector identity to equation (4) with $\overline{A} = \overline{E}$ and $\overline{B} = \overline{H}$,

$$\overline{H} \cdot (\nabla \times \overline{E}) - \nabla \cdot (\overline{E} \times \overline{H}) = \overline{E} \cdot (\sigma \overline{E}) + \overline{E} \cdot \left( \varepsilon \frac{\partial \overline{E}}{\partial t} \right)$$

$$\therefore \quad \overline{H} \cdot (\nabla \times \overline{E}) - \nabla \cdot (\overline{E} \times \overline{H}) = \sigma E^2 + \overline{E} \cdot \left( \varepsilon \frac{\partial \overline{E}}{\partial t} \right) \quad \text{... (4)}$$

Consider first term on left of equation (5). Putting value of $\nabla \times \overline{E}$ from equation (1) we can write,

$$\overline{H} \cdot (\nabla \times \overline{E}) = \overline{H} \cdot \left( -\mu \frac{\partial \overline{H}}{\partial t} \right) = -\mu \overline{H} \cdot \frac{\partial \overline{H}}{\partial t} \quad \text{... (i)}$$

Now consider term,

$$\frac{\partial}{\partial t}(\bar{H}\cdot\bar{H}) = \bar{H}\cdot\frac{\partial\bar{H}}{\partial t}+\bar{H}\cdot\frac{\partial\bar{H}}{\partial t}$$

$$\therefore \quad \frac{\partial}{\partial t}H^2 = 2\bar{H}\cdot\frac{\partial\bar{H}}{\partial t}$$

$$\therefore \quad \frac{1}{2}\frac{\partial}{\partial t}(H^2) = \bar{H}\cdot\frac{\partial\bar{H}}{\partial t} \qquad \text{... (ii)}$$

Similarly we can write,

$$\frac{1}{2}\frac{\partial}{\partial t}(E^2) = \bar{E}\cdot\frac{\partial\bar{E}}{\partial t} \qquad \text{... (iii)}$$

Using results obtained in equations (i), (ii) and (iii) in equation (4),

$$-\frac{\mu}{2}\frac{\partial}{\partial t}(H^2)-\nabla\cdot(\bar{E}\times\bar{H}) = \sigma E^2+\frac{\varepsilon}{2}\frac{\partial}{\partial t}(E^2)$$

$$\therefore \quad -\nabla\cdot(\bar{E}\times\bar{H}) = \sigma E^2+\frac{1}{2}\frac{\partial}{\partial t}[\mu H^2+\varepsilon E^2]$$

But $\bar{E}\times\bar{H}$ is nothing but Poynting vector; $\bar{P}$, rewriting equation,

$$\therefore \quad -\nabla\cdot\bar{P} = \sigma E^2+\frac{1}{2}\frac{\partial}{\partial t}[\mu H^2+\varepsilon E^2] \qquad \text{... (5)}$$

Equation (6) represents **Poynting theorem in point form**. If we integrate this power over a volume, we get energy distribution as,

$$-\int_v \nabla\cdot\bar{P}\,dv = \int_v \sigma E^2\,dv+\frac{\partial}{\partial t}\int_v \frac{1}{2}[\mu H^2+\varepsilon E^2]\,dv$$

Applying divergence theorem to left of above equation, we get,

$$-\oint_S \bar{P}\cdot d\bar{S} = \int_v \sigma E^2\,dv+\frac{\partial}{\partial t}\int_v \frac{1}{2}[\mu H^2+\varepsilon E^2]\,dv \qquad \text{... (6)}$$

Equation (6) represents **Poynting theorem in integral form.** The negative sign on the left of equation (6) indicates that the power is flowing into the surface. The first term on the right gives the total ohmic power loss within the volume, while the second term represents time rate of increase of total energy stored in the electric and magnetic fields. By the law of conservation of energy, the sum of the two terms on right must be equal to the total power flowing into the volume. Thus the **minus**

**In a lossless medium η=40ΠΙ, $\mu_r$=1 and H= 2cos(wt-z) $a_x$ + 5sin(wt-z) $a_y$ Find $\varepsilon_r$,w and E for the medium.**

Given data

Medium is lossless medium or perfect dielectric medium.

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| | $\eta$=40Π<br>$\mu_r$=1<br>H = 2cos(wt-z) $a_x$ + 5sin(wt-z) $a_y$<br>β =1 rad/m<br>a) Finding $\varepsilon_r$<br>$\eta = \sqrt{\frac{\mu}{\varepsilon}} = \frac{\eta_0}{\sqrt{\varepsilon_r}}$<br>$\sqrt{\varepsilon_r} = \frac{\eta_0}{\eta} = \frac{120\pi}{40\pi} = 3$<br>b) Finding ω<br>$v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu_0\mu_r\varepsilon_0\varepsilon_r}} \frac{1}{\sqrt{4\pi X 10^{-7} X 1 X 8.854 X 10^{-12} X 9}}$<br>$v = \frac{\omega}{\beta}$ $\omega = v\beta$<br>c) Finding E $\eta = \frac{E_m^+}{H_m^+} = -\frac{E_m^-}{H_m^-}$<br>$E_m^+ = \eta H_m^+$<br>$E_m^- = -\eta H_m^-$<br>**E = $E^+_m$cos(wt-z) $a_x$ – $E^-_m$sin(wt-z) $a_y$ V/m**<br>**(Substitute the values)** |

7

**Write short notes on: (i) Total internal reflection (ii) Relation between Standing wave ratio reflection coefficient.**

**(i) Total internal reflection**

Total Reflection

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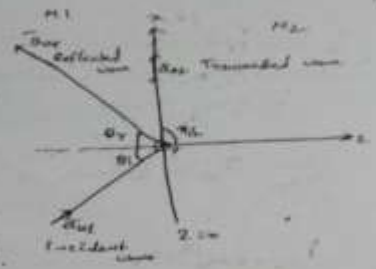

Consider mediums 1 is denser as compared to medium 2 i.e $\varepsilon_1 > \varepsilon_2$. Under this condition the angle of transmission $\theta_t$ becomes greater than angle of incidence. ($\theta_t > \theta_i$).

The angle of transmitted wave increases with the angle of incidence $\theta_i$. when $\theta_t = \pi/2$ transmitted wave will be aligned along the interface as shown in fig.

If $\theta_i$ is increases further, then there will be no transmitted wave and the condition is named as total reflection.

The angle of incidence at which the total reflection takes place is called (critical angle $\theta_c$).

then at $\theta_t = \pi/2$, $\theta_i = \theta_c$

$$\frac{\sin \pi/2}{\sin \theta_c} = \sqrt{\frac{\varepsilon_1}{\varepsilon_2}}$$

$$\sin \theta_c = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$$

$$\theta_c = \sin^{-1}\left(\sqrt{\frac{\varepsilon_2}{\varepsilon_1}}\right) = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$

**(ii) Relation between Standing wave ratio reflection coefficient.**

minimum values of the mag...

It is obvious that when the incident and reflected waves are in phase, ($\phi = 0$) we get max. amplitude of the field. that means the phase angle of both waves is same.

$$\therefore E_{1s}\,max = (1+|\rho|)\,E_i$$

And this maxima occurs where,

$$-\beta_1 z = \beta_1 z + \phi + 2n\pi \qquad \text{where } n = 0, \pm 1, \pm 2, \ldots$$

$$\therefore -\beta_1 z = \frac{\phi}{2} + n\pi$$

Consider that $\eta_1$ & $\eta_2$ both are real and $\eta_2 > \eta_1$, then we get voltage max. at interface (i.e $z=0$) if $\phi = 0$.

If the interface is between a perfect dielectric and

conductor, then $\phi = \pi$.

Then the voltage maximum occurs at

$$-\beta_1 z = \frac{\pi}{2}, \frac{3\pi}{2} \text{ and so on.}$$

when incident wave and reflected wave are out of phase by $180°$, we get minimum voltage.

$$E_{1r\,min} = (1-|\rho|)\,E_i$$

And the minima occurs at where,

$$-\beta_1 z = \beta_1 z + \phi + \pi + 2n\pi \qquad n = 0, \pm 1, \pm 2, \ldots$$

$$\beta_1 z = \frac{\phi}{2} + n\pi + \pi/2$$

from the above equation it is clear that the voltage minima are separated by $\lambda/2$ distance.

thus voltage minima occurs at $-\beta_1 z = 0$ for perfect conductor

" the standing wave ratio (s) is defined as the ratio of maximum & minimum amplitudes of voltages"

$$\boxed{S = \frac{E_{1r\,max}}{E_{1r\,min}} = \frac{1+|\rho|}{1-|\rho|}}$$

i) when the amplitudes of the reflected and incident waves are equal, $|\rho| = 1$, this implies 's' is infinite indicating total energy is reflected.

ii) when there is no reflection of energy i.e $\eta_1 = \eta_2$ then $\rho = 0$, then 's' is unity.

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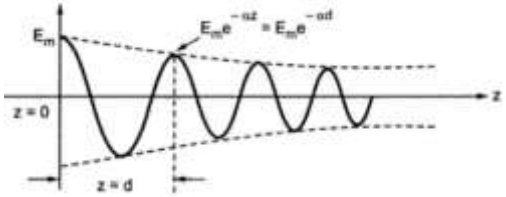
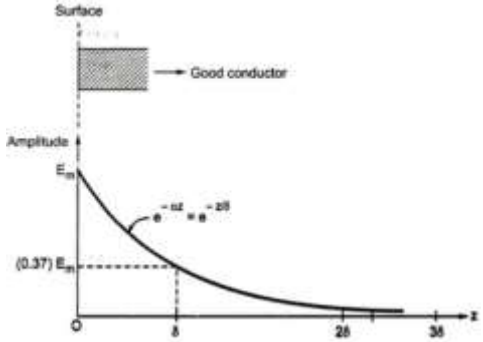
| | 176<br>The standing wave ratio 's' is dimensionless and the values of 's' lies in the range $1 \le s \le \infty$.<br>We can express reflection co-efficient ($\rho$) in terms of the standing wave ratio 's' as,<br>$\boxed{|\rho| = \frac{s-1}{s+1}}$ |
| 8 | **Derive expression for depth of penetration and surface Impedance.**<br>**Depth of penetration**<br>Let the component of electric field Ex travelling in positive z-direction<br>When it travels in good conductor, the conductivity is very high and attenuation constant α is also high. The electric field component in phasor form is<br>$$E_x = E_m^+ e^{-\gamma z} = E_m^+ e^{-\alpha z} e^{-j\beta z}$$<br>When this Ex is propagating in good conductor, there is a large attenuation of amplitude occurs<br>  <br><br>At z = 0, the amplitude of $E_x$ is $E_m$, At z = d the amplitude of $E_x$ is $E_m e^{-\alpha d}$<br>At distance z = d, the amplitude gets reduced by the factor $e^{-\alpha d}$<br>If we select<br>$Z = d = 1/\alpha$<br>Then the reduced factor($e^{-\alpha d}$) becomes<br>$e^{-1} = 0.368$<br>So over the distance d= 1/α, the amplitude of the wave decreases to approximately 37% of its original value.<br>The distance through which the amplitude of travelling wave decreases to 37% of the original amplitude is called as skin depth or depth of penetration. |

It is denoted by δ.




$$SkinDepth = \delta = \frac{1}{\alpha} = \frac{1}{\beta} = \frac{1}{\sqrt{\pi f \mu \sigma}}\, m$$

**Surface Impedance**

to a thin sheet. The **surface impedance** is defined as the ratio of the tangential component of the electric field to the surface current density at the conductor surface. The surface impedance is noted by $Z_s$ and is given by

$$Z_s = \frac{E_{tan}}{J_s}\,\Omega \qquad \text{... (1)}$$

$E_{tan}$ is the tangential component of the electric field which is parallel to the surface of the conductor. The surface current density $J_s$ is the linear current which is the result of the tangential component $E_{tan}$. This indicates the total conduction current per meter width flowing in the thin sheet.

Consider a flat plate conductor with its surface at z = 0 plane, then the current distribution in z-direction is given by,

$$J = J_0\, e^{-\gamma z} \qquad \text{... (2)}$$

where $J_0$ = current density at the surface

$\gamma$ = propagation constant

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| | Note that the thickness of the conductor is assumed to be very large as compared to the depth of penetration of a conductor ; so that the reflection from the back side surface of the conductor is not possible. Then the linear current density is given by,<br>$$J_s = \int_0^\infty J_0 e^{-\gamma z} dz$$<br>$$\therefore \quad J_s = J_0 \left[\frac{e^{-\gamma z}}{-\gamma}\right]_0^\infty$$<br>$$\therefore \quad J_s = \frac{J_0}{\gamma} \quad \text{... (3)}$$<br>But at the surface, the current density is given by,<br>$$J_0 = \sigma E_{tan} \quad \text{... (4)}$$<br>where σ is the conductivity.<br>Hence the linear current density is given by<br>$$J_s = \frac{\sigma}{\gamma} E_{tan} \quad \text{... (5)}$$<br>Substituting values of $J_s$ from equation (5) in equation (1), we get,<br>$$Z_s = \frac{\gamma}{\sigma} \quad \text{... (6)}$$ |
| 9 | **Express surface impedance and all wave characteristics in terms of skin depth.**<br>**Surface Impedance interms of Skin Depth**<br>Surface Impedance is<br>$$Z_s = \frac{E_{tan}}{J_s} \Omega$$<br>$$Z_s = \frac{\gamma}{\sigma} \Omega$$ |

$$Z_s = \frac{\gamma}{\sigma}\Omega$$

$$Z_s = \frac{\alpha + j\beta}{\sigma}$$

$$Z_s = \frac{\alpha}{\sigma} + j\frac{\beta}{\sigma}$$

But Skin depth is

$$\delta = \frac{1}{\alpha} = \frac{1}{\beta}$$

$$Z_s = \frac{1}{\sigma\delta} + j\frac{1}{\sigma\delta}$$

$$Z_s = \frac{1}{\sigma\delta}(1 + j1)$$

$$Z_s = \frac{1}{\sigma\delta}(1 + j1)$$

$$Z_s = \frac{\sqrt{2}}{\sigma\delta}\left(\frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}\right)$$

$$Z_s = \frac{\sqrt{2}}{\sigma\delta}\angle 45^0$$

**Wave Characteristics interms of Skin Depth**

Propagation Constant (γ)

$$\gamma = \alpha + j\beta = \frac{1}{\delta} + j\frac{1}{\delta}$$

Intrinsic Impedance(η)

| | |
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| | $\eta = \sqrt{\frac{\pi f \mu}{\sigma}} + j\sqrt{\frac{\pi f \mu}{\sigma}} = \frac{1}{\sigma\delta} + j\frac{1}{\sigma\delta} = \frac{\sqrt{2}}{\sigma\delta}\angle 45^0$<br>Velocity ($v_p$)<br>$v_p = \frac{\omega}{\beta} = \omega\delta$<br>Wave length ($\lambda$)<br>$\lambda = \frac{2\pi}{\beta} = 2\pi\delta$ |
| 10 | **Define Polarization. Derive the conditions for various polarizations.**<br>**Polarization**<br>The polarization of uniform plane waves is defined as time varying behaviour of the electric field intensity vector $\bar{E}$ at some fixed point in space, along the direction of propagation.<br>Consider a uniform plane wave travelling in positive z-direction. Then the field vectors $\bar{E}$ and $\bar{H}$ lie in x-y plane, which is perpendicular to the direction of propagation of a uniform plane wave. Being an electromagnetic (EM) wave, as wave travels in a space, both the fields under go same variations with respect to time.<br>There are different types of polarization of a uniform plane wave as given below.<br>1) Linear polarization<br>2) Elliptical polarization<br>3) Circular polarization<br>In otherwords, the polarization is nothing but a way in which the magnitude and direction of the electric field varies. These variations of the field are observed in the direction of propagation of a wave. So if the variation of the electric field $\bar{E}$ is observed on a plane perpendicular to the direction of propagation of wave, we get $\bar{E}$ along straight line or circle or ellipse.<br>**Derive the conditions for various polarizations** |

## Conditions for the polarization of sinusoidal waves



Consider that the electric field $\bar{E}$ of a uniform plane wave travelling in z-direction expressed as,

$$\bar{E} = \bar{E}_m e^{-j\beta z}$$

this can be expressed in time varying form as

$$\bar{E} = \bar{E}_m e^{-j\beta z} e^{j\omega t}$$

As the wave is propagating z-direction the normal to xy plane. thus $\bar{E}$ is a resultant of two components $\bar{E}_x$ & $\bar{E}_y$ along x-axis & y-axis respectively.

Let $E_1$ & $E_2$ be a amplitude of $\bar{E}_x$ & $\bar{E}_y$ respectively.

Assume that both the variations are sinusoidal. Let $\delta$ be the phase difference between two components.

the two components can be expressed as in phasor form,

$$\bar{E}_x = \bar{E}_1 e^{j(\omega t - \beta z)}$$

$$\bar{E}_y = \bar{E}_2 e^{j(\omega t - \beta z - \delta)}$$

As the electric field vector $\bar{E}$ is the resultant of $\bar{E}_x$ & $\bar{E}_y$, we can write

$$\bar{E} = \bar{E}_x + \bar{E}_y$$

$$= E_1 e^{j(\omega t - \beta z)} a_x + E_2 e^{j(\omega t - \beta z - \delta)} a_y$$

This equation is true or satisfy for all z values.

for example at z=0

$$\bar{E} = E_1 e^{j\omega t}\,\bar{a}_x + E_2\, e^{j(\omega t-\delta)}\,\bar{a}_y$$

$$\bar{E} = E_1\left[\cos\omega t + j\sin\omega t\right]\bar{a}_x + E_2\left[\cos(\omega t-\delta) + j\sin(\omega t-\delta)\right]\bar{a}_y \quad (1)$$

$$\bar{E} = E_x\,\bar{a}_x + E_y\,\bar{a}_y \quad - (2)$$

Compare real terms of (1) & (2)

$$E_x\,\bar{a}_x + E_y\,\bar{a}_y = \left[E_1\cos\omega t\right]\bar{a}_x + \left[E_2\cos(\omega t-\delta)\right]\bar{a}_y$$

Comparing x & y components both sides

$$E_x = E_1\cos\omega t$$

$$E_y = E_2\cos(\omega t-\delta)$$

$$\frac{E_x}{E_1} = \cos\omega t \Rightarrow \left(\frac{E_x}{E_1}\right)^2 = \cos^2\omega t, \quad 1-\sin^2\omega t$$

$$\frac{E_y}{E_2} = \cos(\omega t-\delta) \qquad \sin^2\omega t = 1-\left(\frac{E_x}{E_1}\right)^2 \Rightarrow \sin\omega t = \sqrt{1-\left(\frac{E_x}{E_1}\right)^2}$$

$$\frac{E_y}{E_2} = \cos\omega t\cos\delta + \sin\omega t\sin\delta$$

Putting the values of $\cos\omega t$ & $\sin\omega t$ in above equation

$$\frac{E_y}{E_2} = \frac{E_x}{E_1}\cos\delta + \sqrt{1-\left(\frac{E_x}{E_1}\right)^2}\,\sin\delta$$

$$\frac{E_y}{E_x} = \frac{E_y}{E_x}\cos\delta + \sqrt{1-\left(\frac{E_x}{E_y}\right)^2}\,\sin\delta$$

Squaring both sides

$$\left(\frac{E_y}{E_x}\right)^2 + \left(\frac{E_x}{E_y}\right)^2\cos^2\delta = \left[1-\left(\frac{E_x}{E_y}\right)^2\right]\sin^2\delta - 2\left(\frac{E_y}{E_x}\right)\left(\frac{E_x}{E_y}\right)\cos\delta$$

$$\left(\frac{E_y}{E_x}\right)^2 + \left(\frac{E_x}{E_y}\right)^2(\cos^2\delta + \sin^2\delta) - 2\left(\frac{E_y}{E_x}\right)\left(\frac{E_x}{E_y}\right)\cos\delta = \sin^2\delta$$

$$\left(\frac{E_x}{E_y}\right)^2 + \left(\frac{E_y}{E_x}\right)^2 - 2\left(\frac{E_x}{E_y}\right)\left(\frac{E_y}{E_x}\right)\cos\delta = \sin^2\delta$$

Above equation is the equation for polarization of monochromatic wave.

Now Apply different conditions

Case(1) $\bar{E}_x$ & $\bar{E}_y$ components are same in phase i.e. $\delta = 0$

from Eqn. $E_x = \left(\frac{E_x}{E_y}\right)E_y$ constant

i.e. we have

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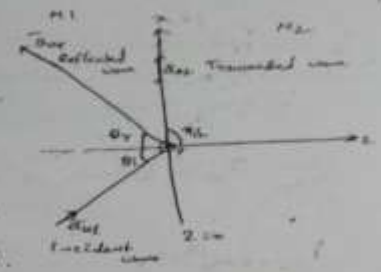
| | Case II) $\bar{E}_x$ & $\bar{E}_y$ components are equal amplitudes with a phase difference $\delta \neq 0$. Let us assume $\delta = \frac{\pi}{2}$<br>$\left(\frac{E_x}{E_1}\right)^2 + \left(\frac{E_y}{E_2}\right)^2 = 1$<br>'ellipse'<br>$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$<br>Case III) $\bar{E}_x$ & $\bar{E}_y$ components have equal amplitudes, with phase difference $\delta = \pi/2$<br>$(\bar{E}_x) = (\bar{E}_y) = E_1 = E_2 = E_0$<br>$\frac{E_x^2}{E_0^2} + \frac{E_y^2}{E_0^2} = 1$<br>$E_x^2 + E_y^2 = E_0^2 \Rightarrow$ 'Circle'<br>$x^2 + y^2 = a^2$ |
| 11 | **Write short notes on:**<br>**(i) Critical Angle. (ii) Brewster angle**<br>**Critical Angle**. |

# Total Reflection




Consider medium 1 is denser as compared to medium 2 i.e $\varepsilon_1 > \varepsilon_2$. Under this condition the angle of transmission $\theta_t$ becomes greater than angle of incidence. ($\theta_t > \theta_i$).

The angle of transmitted wave increases with the angle of incidence $\theta_i$. when $\theta_t = \pi/2$ transmitted wave will be aligned along the interface as shown in fig.

If $\theta_i$ is increases further, then there will be no transmitted wave and the condition is named as total reflection.

The angle of incidence at which the total reflection takes place is called (critical angle $\theta_c$).

Then at $\theta_t = \pi/2$, $\theta_i = \theta_c$

$$\frac{\sin \pi/2}{\sin \theta_c} = \sqrt{\frac{\varepsilon_1}{\varepsilon_2}}$$

$$\sin \theta_c = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$$

$$\theta_c = \sin^{-1}\left(\sqrt{\frac{\varepsilon_2}{\varepsilon_1}}\right) = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$